STRUCTRA
TRAINING

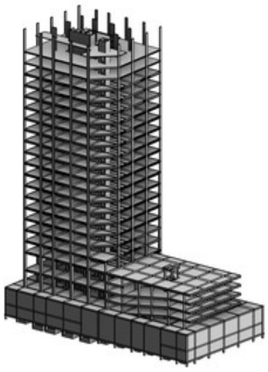
SEPTEMBER 2026 INTAKE

AUTODESK REVIT / STRUCTURAL BIM TRAINING

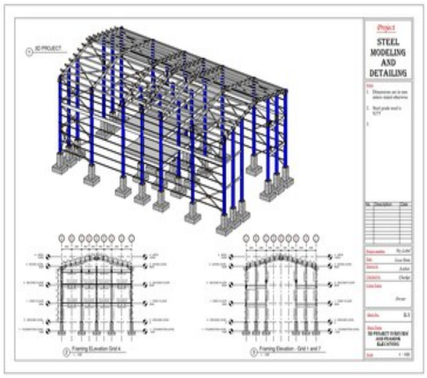
REVIT FOR STRUCTURES

Build coordinated, buildable structural models — from datum to documentation.

REINFORCED CONCRETE TRACK   STEEL TRACK



CONCRETE STRUCTURES



STEEL STRUCTURES

PROGRAM OVERVIEW

Revit for Structures is a hands-on program for engineering students and early-career professionals who want to build real structural models — not just attractive renders. Choose either focused track or take both, progressing from Revit fundamentals through modeling, detailing, schedules and construction documentation.

START DATE
Monday, 14 September 2026
Third week of September

FORMAT
100% online
Live instructor-led sessions via Zoom

DURATION
4 weeks per track
Weekday evenings

FEE
KSH 2,500 per track

WHO IT'S FOR
University engineering students and early-career structural / BIM professionals

PREREQUISITES
A laptop capable of running Revit; no prior Revit experience required

TRACK 01 / STRUCTURAL MODELING + REINFORCEMENT

REINFORCED CONCRETE STRUCTURES

BIM

01 Brief overview of Revit

- Difference with AutoCAD software
- Benefits over AutoCAD software

02 Setting up the software

- Brief introduction of libraries required
- Downloading other extensions from the Autodesk site

03 Beginning with Revit

- Creating a new project
- An overview of the templates
- An overview of palettes, tabs and panels

04 Creating datum elements

- Creating levels
- Creating grids
- Propagating grid extents
- Customizing grid bubble shape and appearance
- Linking CAD files to create grids

05 Overview of creating 3D elements

- Foundations, columns, beams and slabs

06 Creating columns

- Placement of columns
- Loading column families

07 Creating foundations

- Hosting foundations on the columns created
- Creating strip footing
- Using the eccentricity option
- Basement floor slab
- Loading foundation families into your drawings

08 Creating beams and slabs

- Copying beams and slabs from level to level
- Creating a slab
- Hollow pot and waffle slabs

09 Curved beams

- Procedure for modeling curved beams

10 Beam system layout and justification rules

- Using the beam system to model structural beams
- Modeling beams using beam system layout rules
- Modeling beams using beam system justification rules
- Deselecting the beam system

11 Creating staircases and ramps

- Types of staircases
- Creating an architectural ramp and a structural ramp
- Modeling a staircase as a ramp
- Shaft openings

TRACK 01 / STRUCTURAL MODELING + REINFORCEMENT

REINFORCED CONCRETE STRUCTURES

BIM

12 Visibility/graphics and filters

- Applying visibility/graphic overrides — method 1
- Applying visibility/graphic overrides — method 2
- Applying visibility/graphic overrides — method 3
- Applying visibility/graphic overrides — method 4

13 Creating roofs and truss placement

- Sloped and gable-ended roofing
- Adding system trusses to your roof
- Editing and creating customized trusses

14 Placing reinforcement

- Creating sections in your drawing
- Foundation reinforcement
- Column reinforcement
- Beam reinforcement
- Slab reinforcement
- Staircase reinforcement
- Ramp reinforcement

15 Annotations

- Adding dimensions
- Adding spot elevations
- Adding spot coordinates
- Adding spot slope

16 Detailing of sections

- Placing breaklines in sections
- Using hatch files to create regions
- Placement of rebar tags on reinforcement
- Adding comments on reinforcement
- Tagging structural elements

17 Schedules and material takeoffs

- Creating bar bending schedules
- Loading shape images into bar bending schedules
- Creating material takeoffs for structural elements
- Sorting and filtering fields in schedules

18 Placing drawings on sheets

- Loading different paper sizes for title blocks
- Using system-loaded title blocks
- Editing and customizing title blocks
- Adjusting text and fields in a title block
- Saving title blocks as a Revit family template
- Duplicate options for drawings
- Placing drawings on sheets

19 Printing sheets to PDF format

- Setting up sheets before printing
- Printing one sheet at a time
- Printing several sheets at a time

20 Converting files into CAD and other formats

- Export options for your drawing
- Transporting your drawing to AutoCAD
- Sending your model for analysis

TRACK 02 / MODELING + CONNECTION DETAILING

STEEL STRUCTURES

BIM

01 Brief overview of Revit

- Difference with CAD software
- Benefits over CAD software

02 Setting up the software

- Brief introduction of libraries required
- Downloading other extensions from the Autodesk site

03 Beginning with Revit

- Creating a new project
- An overview of the templates
- An overview of palettes, tabs and panels

04 Creating datum elements

- Creating levels
- Creating grids
- Elevation markers
- Propagating grid extents
- Customizing grid bubble shape and appearance (optional)
- Linking CAD files to create grids

05 Modeling steel columns

- Placement of columns
- Loading column families

06 Modeling concrete columns

- Placement of concrete columns
- Loading column families

07 Modeling the foundations

- Hosting foundations on the columns created
- Loading other foundation families into your drawings

08 Modeling the rafters

- Placement of rafters
- Loading steel beam sections
- Attaching columns to the rafters
- Copying rafters from grid to grid

09 Modeling the steel beams

- Loading steel beam sections
- Modeling the floor beams
- Justification rule of the beams
- Copying the floor beams
- Modeling beams at the eaves level

10 Beam system layout and justification rules

- Using the beam system to model structural beams
- Modeling beams using beam system layout rules
- Modeling beams using beam system justification rules
- Deselecting the beam system

TRACK 02 / MODELING + CONNECTION DETAILING

STEEL STRUCTURES

BIM

11 Composite slab and composite beam

- Modeling the composite beam
- Modeling the composite slab
- Cantilevering the composite slab

12 Floor openings

- Using the shaft opening tool

13 Curved beams

- Procedure for modeling curved beams

14 Sloped / slanted beams

- Modeling sloped beams using the offset tool
- Modeling sloped beams using the framing elevation

15 Purlins and side rails

- Modeling purlins using reference planes
- Modeling purlins using the beam system
- Modeling the side rails

16 Bracings

- Vertical bracings
- Horizontal bracing

17 Visibility/graphics and filters

- Applying visibility/graphic overrides — method 1
- Applying visibility/graphic overrides — method 2
- Applying visibility/graphic overrides — method 3
- Applying visibility/graphic overrides — method 4
- Hiding elements

18 Creating sections

- Types of sections
- Creating sections in your drawing

19 Placing reinforcement

- Setting up the reinforcement cover
- Placing reinforcement at the foundation
- Column reinforcement
- Slab reinforcement

20 Splitting elements

- Splitting elements

21 Introduction to connections

- Uploading the connections
- Applying connections to structural elements
- Propagating the connections

TRACK 02 / MODELING + CONNECTION DETAILING

STEEL STRUCTURES

BIM

22 Connection types

- Main types of connection
- Haunch connection
- Apex connection
- Base plate connection
- Gusset plate bracing connection
- Gusset plate column-base plate connection
- Purlin / side rail connection
- Single-sided end plate (simple connection)
- Single-sided end plate (moment connection)
- Double-sided end plate (simple connection)
- Fin plate (simple connection)
- Clip angle (simple connection)
- Column splice connection
- Creating custom connections

23 Fabrication elements, modifiers and parametric cuts

- Use and location

24 Annotations

- Adding dimensions
- Adding spot elevations
- Adding spot coordinates
- Adding spot slope
- Creating custom bolt and plate tags

25 Detailing of sections

- Placing breaklines in sections
- Using hatch files to create regions
- Placement of rebar tags on reinforcement
- Adding comments on reinforcement
- Tagging structural elements

26 Schedules and material takeoffs

- Creating bar bending schedules
- Loading shape images into bar bending schedules
- Creating material takeoffs for structural elements
- Sorting and filtering fields in schedules

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- Transporting your drawing to AutoCAD
- Sending your model for analysis